# A BIJECTIVE CONTINUOUS MAP FROM A QUASI-COMPACT TO A HAUSDORFF SPACE IS A HOMEOMORPHISM

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-topology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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*Editorial note.* Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

*Allowed standard prerequisites (editorial).*

- The statement that a closed subset of a quasi-compact space is quasi-compact. Invoked inside the counted proof (source line 3506). Makes images of closed subsets quasi-compact.
- The statement that the image of a quasi-compact set under a continuous map is quasi-compact. Invoked inside the counted proof (source line 3507). Keeps quasi-compactness along the map.
- The statement that a quasi-compact subset of a Hausdorff space is closed. Invoked at the end of the counted proof (source line 3508). Concludes closedness of the images.

*Source reference.* Original lemma lemma-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1879–1879. Navigation pointer only: the counted text refers to the Stacks item lemma-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

*Source reference.* Original lemma lemma-image-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1964–1964. Navigation pointer only: the counted text refers to the Stacks item lemma-image-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

*Source reference.* Original lemma lemma-quasi-compact-in-Hausdorff: sources/stacks-topology/pinned-original-source.tex, source lines 1894–1894. Navigation pointer only: the counted text refers to the Stacks item lemma-quasi-compact-in-Hausdorff,

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which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

**Lemma 0.1.** *Let $f : X \to Y$ be a continuous map of topological spaces. If $f$ is bijective, $X$ is quasi-compact, and $Y$ is Hausdorff, then $f$ is a homeomorphism.*

*Proof.* It suffices to prove $f$ is closed, because this implies that $f^{-1}$ is continuous. If $T \subset X$ is closed, then $T$ is quasi-compact by Lemma lemma-closed-in-quasi-compact, hence $f(T)$ is quasi-compact by Lemma lemma-image-quasi-compact, hence $f(T)$ is closed by Lemma lemma-quasi-compact-in-Hausdorff. □